Accuracy$^{\text{Coverage}}$ (%)

Dataset	MedGemma									Qwen									NAR					
	LL			Head			Adapt			LL			Head			Adapt			MedGemma					
	Base	+ Conformal $\alpha = 0.3$	+ Routing $\tau = 0.7$	Base	+ Conformal $\alpha = 0.2$	+ Routing $\tau = 0.5$	Base	+ Conformal $\alpha = 0.1$	+ Routing $\tau = 0.7$	Base	+ Conformal $\alpha = 0.3$	+ Routing $\tau = 0.5$	Base	+ Conformal $\alpha = 0.3$	+ Routing $\tau = 0.5$	Base	+ Conformal $\alpha = 0.2$	+ Routing $\tau = 0.7$	Base	+ Conformal $\alpha = 0.3$	+ Routing $\tau = 0.3$
PubMedQA	71.2	**75.5**$^{88.8}$	72.5$^{97.6}$	65.6	**71.9**$^{74.0}$	71.0$^{77.8}$	72.6	**75.6**$^{89.2}$	73.0$^{97.0}$	68.2	**71.5**$^{82.8}$	71.1$^{92.2}$	65.6	69.9$^{89.0}$	**70.7**$^{79.2}$	68.6	**72.4**$^{90.0}$	71.3$^{92.0}$	64.0	**64.3**$^{99.6}$	64.0$^{100.0}$
MedQA	49.4	**56.4**$^{64.7}$	50.8$^{93.1}$	51.5	62.3$^{53.7}$	**65.9**$^{48.7}$	53.3	**63.5**$^{60.3}$	57.1$^{84.0}$	43.8	48.0$^{64.4}$	**50.2**$^{63.3}$	44.9	51.1$^{64.2}$	**63.0**$^{27.5}$	45.4	**54.0**$^{52.0}$	53.9$^{55.3}$	30.4	**31.7**$^{29.3}$	–$^{0.0}$
MedMCQA	60.1	**65.8**$^{78.0}$	60.9$^{97.0}$	65.5	**76.7**$^{65.8}$	76.6$^{67.3}$	68.7	**91.5**$^{37.2}$	81.7$^{67.2}$	52.9	46.8$^{51.3}$	**62.8**$^{63.1}$	55.9	62.7$^{75.2}$	**73.9**$^{49.6}$	56.4	**68.6**$^{61.4}$	66.3$^{66.5}$	40.8	54.6$^{27.0}$	**78.9**$^{2.9}$
Text Average	60.2	**66.9**$^{77.2}$	61.6$^{95.9}$	60.8	70.9$^{64.5}$	**71.6**$^{64.6}$	64.9	**74.9**$^{62.2}$	70.0$^{82.7}$	55.0	57.5$^{66.2}$	**62.7**$^{72.9}$	55.4	62.2$^{76.1}$	**70.4**$^{52.1}$	56.8	**66.6**$^{67.8}$	65.3$^{71.3}$	45.1	56.5$^{52.0}$	**64.4**$^{34.3}$
VQA-RAD	70.6	**77.4**$^{85.3}$	71.3$^{99.1}$	**76.1**	76.1$^{80.7}$	**76.1**$^{100.0}$	69.7	**74.5**$^{89.9}$	70.4$^{99.1}$	65.1	**74.4**$^{75.2}$	65.1$^{100.0}$	70.6	**75.3**$^{85.3}$	70.6$^{100.0}$	66.1	**71.6**$^{80.7}$	69.1$^{86.2}$	56.0	**56.7**$^{95.4}$	56.0$^{100.0}$
SLAKE	79.4	**82.4**$^{86.5}$	80.1$^{97.5}$	86.8	**87.7**$^{96.1}$	86.8$^{100.0}$	90.7	**92.6**$^{95.8}$	91.6$^{97.5}$	71.3	**75.7**$^{85.9}$	71.3$^{100.0}$	78.3	**80.1**$^{93.5}$	78.3$^{100.0}$	85.1	**89.0**$^{92.1}$	86.5$^{96.3}$	80.0	**80.4**$^{99.2}$	80.0$^{100.0}$
PathVQA	62.4	**67.1**$^{70.4}$	63.1$^{95.8}$	86.3	**89.2**$^{91.4}$	86.3$^{100.0}$	90.0	**93.9**$^{88.0}$	91.0$^{96.8}$	61.9	**65.6**$^{77.1}$	61.9$^{100.0}$	84.8	**85.7**$^{96.2}$	84.8$^{100.0}$	86.3	**90.0**$^{89.2}$	88.3$^{93.3}$	86.9	**87.1**$^{99.0}$	86.9$^{100.0}$
PMC-VQA	48.9	**53.8**$^{75.2}$	49.6$^{96.9}$	57.1	68.3$^{58.4}$	**68.7**$^{57.9}$	57.2	**71.2**$^{46.5}$	61.5$^{78.8}$	49.1	**56.4**$^{60.4}$	56.2$^{63.6}$	56.7	62.6$^{80.1}$	**71.1**$^{54.1}$	57.3	**65.8**$^{65.2}$	64.8$^{70.2}$	45.4	49.4$^{71.7}$	**50.8**$^{65.9}$
Image Average	65.4	**70.9**$^{79.3}$	66.1$^{97.3}$	76.6	**81.8**$^{81.6}$	80.7$^{89.5}$	76.9	**84.8**$^{80.0}$	79.4$^{93.0}$	61.8	**68.9**$^{74.7}$	64.3$^{90.9}$	72.6	76.5$^{88.8}$	**76.9**$^{88.5}$	73.7	**80.4**$^{81.8}$	78.3$^{86.5}$	67.0	69.9$^{91.3}$	**70.0**$^{91.5}$
Overall Average	63.1	**69.2**$^{78.4}$	64.2$^{96.7}$	69.8	**77.7**$^{74.3}$	77.5$^{78.8}$	71.8	**81.1**$^{72.4}$	75.6$^{88.6}$	58.9	**64.3**$^{71.0}$	63.7$^{83.2}$	65.2	70.9$^{83.4}$	**74.9**$^{72.9}$	66.5	**75.1**$^{75.8}$	73.3$^{80.0}$	57.6	65.9$^{74.5}$	**68.8**$^{67.0}$